PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

umhackathon@um.edu.my

© UMHackathon 2026

Email: umhackathon@um.edu.my

Table of Contents

1. Project Overview 1
2. Background & Business Objective 1
3. Product Purpose 2
4. System Functionalities 2
5. User Stories & Use Cases 3
6. Features Included (Scope Definition) 3
7. Features Not Included (Scope Control) 3
8. Assumptions & Constraints 4
9. Risks & Questions Throughout Development 4

© UMHackathon 2026

Email: umhackathon@um.edu.my

Project Name: Z Intelligence

Version: 1.0

1. Project Overview

- **Problem Statement:** Many Malaysian SME e-commerce sellers struggle with fragmented data across platforms like Shopee, Lazada, and TikTok Shop, leading to manual, intuition-based decision-making that misses critical revenue opportunities. Existing tools focus on simple automation rather than complex "What-if" reasoning, leaving non-technical business owners without the specialized analytical skills required to visualize growth strategies or identify hidden operational risks.
- **Target Domain:** Enabling AI-powered Economic Empowerment and Decision Intelligence for SME e-commerce sellers by unifying scattered data into actionable business strategies.
- **Proposed Solution Summary: Z Intelligence** acts as a unified "Command Center" that removes the friction of managing multi-platform data by enabling sellers to interact with a guided three-agent GLM architecture in simple English. By shifting the user's role from a manual data builder to a strategic **Curator**, the system provides instant risk detection (Risk Sentinel), business analysis (Strategist), and automated communication drafts (Executor) to optimize measurable economic outcomes like revenue improvement and cost reduction.

© UMHackathon 2026

Email: umhackathon@um.edu.my

2. Background & Business Objective

- **Background of the Problem:** For many years, the management of multi-platform e-commerce businesses followed a manual and fragmented pathway. A business owner would log into individual seller centers for Shopee, Lazada, and TikTok Shop, manually aggregate sales and inventory data into spreadsheets, and rely on intuition to determine pricing or restock levels. This process requires extensive labor, specialized analytical skills, and significant time, blocking SME owners from effectively visualizing growth opportunities or identifying hidden operational risks.
- **Importance of Solving This Issue:** By accelerating the time required to transform scattered data into actionable information from days to seconds, SME owners can test business strategies and trade-off scenarios at an unprecedented rate. Breaking the technical skill barrier for advanced data analytics allows anyone to visualize their business status through a "Command Center" without years of training in business intelligence. This shift not only optimizes productivity but directly influences economic empowerment by increasing revenue and reducing operational costs through AI-driven precision.

- **Strategic Fit / Impact:**

  Z Intelligence directly addresses the AI for Economic Empowerment framework by transforming Z.AI's GLM from a generative tool into a central reasoning engine for SME operational growth. By integrating GLM into a specialized three-agent architecture — Strategist, Risk Sentinel, and Executor — via CrewAI, the system moves beyond data visualization into Autonomous Decision Intelligence.

  The project addresses a critical and measurable gap: only 54% of Malaysian SMEs currently use data analytics The Malaysian Reserve despite SMEs contributing 39.1% of Malaysia's GDP BERNAMA, creating a systematic blind spot in business

© UMHackathon 2026

decision-making. Z Intelligence bridges this gap through an AI-native lifecycle — ingesting fragmented data from multi-platform e-commerce sources into Firebase, computing 22 pre-aggregated business metrics, and providing sellers a direct conversational channel with the Strategist agent for real-time clarification and scenario planning.

The projected impact is grounded in published research: AI-driven pricing optimization has demonstrated up to 14.2% sales growth and 23.9% margin increases in retail deployments Unikie, while organizations adopting AI analytics report 5–6% increases in productivity translating directly to improved profit margins Psico-smart. For supplier communication specifically, AI adoption in sales and procurement reduced deal closure times by 25% DRING AI — directly validating the Executor agent's automated supplier drafting capability. With SMEs representing 98.5% of Malaysian business establishments The Malaysian Reserve, Z Intelligence provides a scalable, realistic, and measurable solution for the digital empowerment of the Malaysian SME economy.

3. Product Purpose

3.1. Main Goal of the System: To provide an AI-enabled Decision Intelligence (DI) partner that empowers SME owners to shift from manual data builders to strategic curators of their business growth. By unifying fragmented platform data and utilizing a guided multi-agent architecture, the system transforms raw information into actionable strategies that increase revenue and reduce operational costs.

3.2. Intended Users (Target Audience):

- Product Manager
- Entrepreneurs / Founders
- E-commerce SME Sellers
- Digital Strategists

© UMHackathon 2026

- Inventory Managers
- Marketing Managers

4. System Functionalities

4.1. Description: The system operates as a high-speed **Decision Intelligence (DI) engine** that transforms fragmented e-commerce data into actionable business strategies through natural language. Using an **Autonomous Multi-Agent Orchestration** model, the system captures complex business relationships, identifies operational risks, and projects financial outcomes (revenue, profit, and margin) to facilitate data-driven leadership for SME owners.

4.2. Key Functionalities:

- **Risk Sentinel Monitoring**: Translates raw data triggers into severity-coded alerts, automatically ranking business threats like stockouts or margin erosion to focus the owner's attention on critical issues.

- **Conversational Strategy Analysis**: Facilitates continuous modification of business scenarios (e.g., "What if I increase price?") by allowing users to query the Strategist Agent for immediate predictive insights and trade-off comparisons.

- **Guided Action Execution**: Enables the conversion of AI recommendations into professional, component-based drafts for supplier emails or WhatsApp messages, bridging the gap between "insight" and "action."

- **Unified Data Extract**: Automatically merges scattered data from multi-platform sources (E-commerce platforms APIs and Unstructured data from user uploads) into a single, time-stamped Business Context DB, maintaining a "versioned" history of business performance.

© UMHackathon 2026

- **Report Generation:** Getting the information alone is not enough, having the function to generate reports to allow discussion among the team is important as well. Z-Intelligence allows the user to prompt the Executor Agent to generate reports according to the user's needs, selecting only what is required, and getting the report in their desired format.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Z.AI's ilmu-glm-5.1 (General Language Model). The ilmu-glm-5.1 was chosen for its superior **multi-step reasoning** capabilities and its high proficiency in the **Malaysian e-commerce context**. Its ability to interpret both structured pre-aggregated KPIs (from Firebase) and unstructured data (competitor prices) makes it the ideal core engine for handling complex trade-off analyses that require a deep understanding of SME business logic.

4.3.2. Prompting Strategy: The core strategy is a multi-agent ReAct pipeline (via CrewAI) with heavy emphasis on structured output enforcement and context injection to reduce hallucination and unnecessary API calls. Multi-step Agentic Prompting using a "Guided Agent" architecture. This chosen strategy fits our use case because e-commerce decisions are too multi-faceted for a single prompt. We split complex queries into sub-tasks delegated to specialized agents (Sentinel, Strategist, Executor). By using **"Agent-as-a-Tool"** logic, we ensure that each GLM call is highly focused on a specific data subset, significantly reducing "hallucinations" and increasing the precision of strategic advice.

4.3.3. Context & Input Handling: The system uses a Pre-aggregation Engine to summarize raw platform data into key performance indicators (KPIs) before it reaches the GLM. The agent-level constraint is max_iter (maximum iterations per agent, set to 1–3), which is important to limit reasoning loops and slow response. Additionally, the backend selects only the

© UMHackathon 2026

top 5 relevant KPIs for the specific insight category (e.g., top 5 best-selling products) to ensure the prompt remains focused and high-quality.

4.3.4. Fallback & Failure Behavior: For critical actions, a manual validation safeguard requires the user to review and edit any AI-drafted communications before use. Additionally, an Arithmetic Integrity Check compares AI projections against pre-aggregated historical bounds; if a discrepancy is detected (potential hallucination), the projection is disabled and the user is guided to the manual data workspace for verification.

5. User Stories & Use Cases

- **User Stories:**
  - "As an SME seller operating on Shopee and Lazada, I want to view my net profit, revenue, and margin across all platforms in a single dashboard so that I do not need to log into multiple seller centres to understand whether my business is actually making money."
  - "As a business owner, I want to type a question like 'Why did my revenue drop this week?' in plain language so that Z-Intelligence analyses my KPIs and tells me which products, platforms, or campaigns are responsible — without me building a single spreadsheet."
  - "As an SME seller managing dozens of SKUs, I want to receive a P1 critical alert automatically when any product's stock falls below 7 days of supply so that I can restock before a stockout during a peak sale event like 11.11 or 12.12."
  - "As an operations manager, I want to monitor the return rate per product and per category weekly so that I can identify quality or logistics issues before they damage my seller rating and erode profit."
  - "As a campaign manager, I want to see the ROI, ROAS, and Margin Impact of every active voucher campaign so that I can immediately pause loss-making promotions and increase budget on those that generate real profit."

© UMHackathon 2026

- "As a first-time e-commerce seller with no data analytics background, I want Z-Intelligence to explain its recommendations in plain, jargon-free language so that I can act on the insight without needing to understand the underlying formula."

- **Use Cases (Main Interactions):**

  - **Natural Language Query to AI Insight Flow:** The seller types a plain-language business question into the Z-Intelligence chat interface (e.g., "Which SKUs should I restock this week?" or "Is my Shopee voucher campaign profitable?"). Z-Intelligence retrieves the relevant pre-aggregated metrics from DS7, constructs a structured prompt containing KPI context alongside the seller's query, and sends it to Z.AI GLM. The GLM returns a context-aware, plain-language recommendation with the specific metrics that informed it (e.g., "SKU A004 has 5 inventory days remaining vs. a 14-day restock lead time — recommend ordering 200 units"). The insight is rendered in the chat interface alongside the referenced KPI values for transparency.

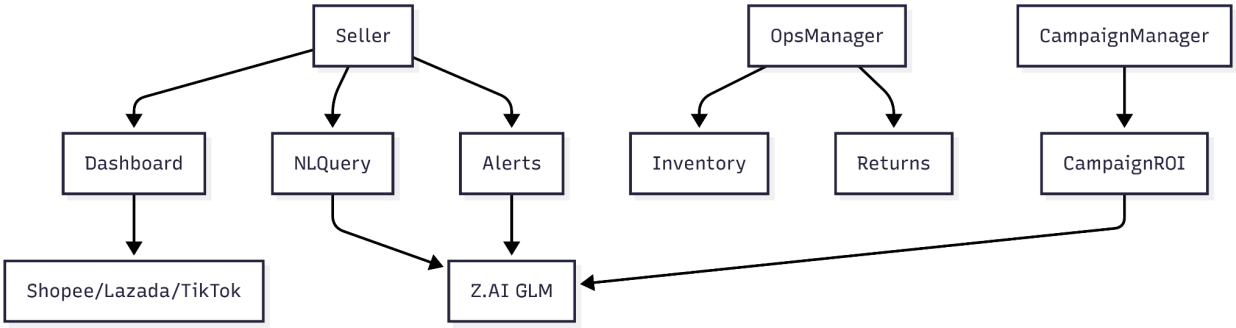
  - **Automated Alert Detection and Notification Flow:** The Alert Manager continuously monitors DS7 for threshold breaches across all 28 pre-aggregated KPIs. When Inventory Days Remaining for any SKU drops to ≤ 7 days, or when the Return Rate surges above the seller's configured threshold, or when a supplier COGS upload shows a >10% cost increase, the Alert Manager classifies the event as P1 (Critical) or P2 (Warning) and pushes a real-time notification badge to the seller's dashboard. The seller taps the alert to open the detail view, where the GLM has already generated a recommended remediation action (e.g., "Your margin on SKU B012 dropped 14% this week due to a shipping fee increase — consider adjusting the Lazada listing price by RM 2.50 to restore margin above 20%").

© UMHackathon 2026

- **Multi-Platform Data Sync and Dashboard Rendering:** The seller connects their Shopee, Lazada, TikTok Shop, and/or Shopify API credentials in the onboarding flow. Z-Intelligence's Data Ingestion & Normalisation process pulls raw orders, inventory snapshots, campaigns, ad spend, and customer data via each platform's API. The data is validated, deduplicated, and transformed into the 3NF schema across DS1–DS6. The Pre-Aggregation Engine then computes all 22 KPI metrics and writes results to DS7. The Dashboard Renderer reads DS7 to display the 6 primary KPI cards — Net Profit, Net Margin %, Inventory Days Remaining, Return Rate %, WoW Revenue Change, and Top 3 Products by Net Profit — refreshed on a daily schedule or triggered on-demand by the seller.

- **Campaign Performance Analysis Flow:** The seller navigates to the Campaign tab and selects an active voucher or ad campaign. Z-Intelligence fetches Campaign ROI, ROAS, Voucher Usage Rate, and Margin Impact of Voucher from DS7 for the selected campaign. The GLM contextualises the performance (e.g., comparing ROAS against the seller's historical average and flagging if a campaign's voucher discount is cannibalising gross margin). The output is a plain-language verdict: scale, pause, or modify — with the specific metric values supporting the call.

- **Use Case Diagram:**



© UMHackathon 2026
Email: umhackathon@um.edu.my

6. Features Included (Scope Definition)

- Centralized Multi-Platform Interface: A unified web dashboard that aggregates "scattered data" from multiple different E-commerce platforms such as Shopee, Lazada, and TikTok Shop into a single source of truth using Firebase Firestore.

- Conversational AI Strategist Agent: A text-based interface powered by ilmu-glm-5.1 that allows users to prompt for "What-if" simulations, revenue projections, and pricing trade-off analysis.

- Autonomous Risk Sentinel: A background monitoring system that detects operational anomalies (stockouts, margin drops, high return rates), ranks them by P1/P2 severity, and alerts the user in real-time.

- Multi-Modal Data Extraction (Z-OCR & Voice): A functional extraction workspace that standardizes unstructured inputs—specifically Receipt photos, PDF invoices, and Voice Notes—into the business database with real-time confidence scoring.

- One-Click "Quick Action" Executor: A specialized page that transforms AI strategic recommendations into professional, editable drafts for Supplier Emails and Team WhatsApp messages.

© UMHackathon 2026

Email: umhackathon@um.edu.my

- Verified Pre-aggregation Engine: A dedicated backend logic layer that pre-calculates all financial KPIs to ensure 100% arithmetic accuracy before data is presented to the GLM for reasoning.

- Business Impact Tracker: Real-time visualization of the quantifiable value generated by the system, tracking "Estimated Profit Increase" and "Reduction in Return Rates" (Impact Accumulated).

7. Features Not Included (Scope Control)

- Live Marketplace Production OAuth Sync: For the hackathon demo, the system utilizes a Standardized Firebase Dataset to simulate platform responses rather than requiring live Shopee/Lazada seller production credentials.

- Fully Autonomous Execution: Z Intelligence is a "Guided Agent" platform; it generates drafts for review but does not support fully autonomous inventory purchasing or direct message sending without human approval.

- Consumer-Facing Support (B2C): The system is optimized for B2B operations and internal management; it does not host a real-time chat infrastructure for end-consumers or buyers.

- Autonomous Financial Transactions: The system provides decision intelligence and profit projections but does not execute actual payments, bank transfers, or wallet withdrawals.

- Multi-User Collaborative Editing: The current prototype is optimized for a single business owner session and does not support simultaneous multi-user inputs or real-time collaborative dashboard editing.

© UMHackathon 2026

- Long-Term Predictive Modeling: The system focuses on short-term operational optimization (weekly/monthly trends) and does not support multi-year macro-economic forecasting.

8. Assumptions & Constraints

- **LLM Cost Constraint:** We estimate a token cost of approximately 10,000 - 20,000 tokens per average user session, covering risk scanning, strategy simulation, and communication drafting. To keep inference costs manageable, we implemented a "Summarization-First" Backend. Instead of feeding raw platform data into the GLM, our backend pre-aggregates raw JSON into a concise KPI summary (e.g., Top 5 products, Net Margin %, Low Stock alerts). This prevents the model from processing thousands of unnecessary data lines, significantly reducing the input token count.
- **Technical Constraints:** The Executor Agent is currently limited to generating text-based professional templates for Email (SMTP) and WhatsApp. Direct integration with internal ERP systems or legacy inventory software is not supported in the initial version. The system only supports structured data inputs in JSON or CSV format following our predefined schema. Unstructured inputs, such as raw handwritten ledger photos, are not supported for the preliminary round.
- **Performance Constraints:** To maintain high-quality reasoning and avoid GLM "exhaustion" or hallucination, the Strategist Agent is limited to 5 "What-if" simulations per hour per user. High-fidelity strategy generation involves multi-step agentic reasoning, which may result in a processing delay of 5 to 10 seconds per request.

© UMHackathon 2026

- **User Input:** The system is not a fully autonomous "black box"; it cannot generate business strategies without an initial human prompt or a data-triggered alert to define the context. Because the system is in a prototype phase and not yet ready for fully automated production, all high-stakes outputs (like financial projections and supplier communications) require human approval before they are finalized or sent.

9. Risks & Questions Throughout Development

- **Logic Hallucination & Arithmetic Accuracy:** How do we ensure the GLM does not "hallucinate" incorrect profit margins or inventory counts?
    - Mitigation: The system uses a Pre-aggregation Engine to perform all math in the backend. The AI only receives the results of these calculations to interpret, never raw uncalculated numbers.
- **Context Window Overflow:** What happens if a seller has thousands of SKUs that exceed the 800-token limit of the agent's context?
    - Mitigation: We implemented "Summarization-First" logic. The system identifies and sends only the "Top 5 movers" or "Critical 5 risks" to the GLM, rather than the entire inventory list.
- **Multi-Platform Data Reconciliation:** How do we handle discrepancies if the same product has different names or SKUs across Shopee, Lazada, and TikTok?
    - Question: Should the system force a manual mapping by the user in the Data Workspace, or rely on AI-fuzzy matching?

© UMHackathon 2026
Email: umhackathon@um.edu.my

- **Agent Logic Contradiction:** What happens if the Risk Sentinel identifies a need to restock, but the Strategist Agent suggests pausing that product due to low margins?
    - Risk: Conflicting AI advice could confuse the user. We need to define a "Priority Hierarchy" where survival risks (stockouts) outrank margin optimizations.
- **Explainability & Trust (The "Black Box" Risk):** Will SME owners trust a RM0.50 price hike suggestion if they can't see the underlying competitor data?
    - Mitigation: The "View Detailed Analysis" feature is designed specifically to show the "Chain of Thought" and data evidence used by the Strategist Agent.
- **API Rate Limiting & Latency:** E-commerce APIs (Shopee/Lazada) often have strict rate limits. How do we ensure the "Live Sync" remains live without getting blocked?
    - Mitigation: Shopee and Lazada enforce API rate limits that prevent continuous real-time polling. MajuAI addresses this through a scheduled sync architecture: the Data Ingestion process (P1.0) fetches raw data from platform APIs on a configurable interval (default: every 30 minutes), with exponential backoff on rate-limit responses (HTTP 429). All fetched data is persisted to Firestore (DS1–DS6). The UI reads exclusively from Firestore — never directly from platform APIs — ensuring consistent dashboard performance regardless of upstream API availability. The "Last updated" timestamp shown on the dashboard reflects the most recent successful sync cycle.
- **Strategy Repetition & Bias:** How do we ensure the AI doesn't give the same generic "increase marketing" advice to every user regardless of their specific niche?
    - Mitigation: By injecting specific Competitor Benchmarking data into the prompt, ensuring the advice is unique to the user's current market position.
- **Data Privacy & Security:** SME financial data is highly sensitive. How are we securing the Service Account JSON and the user's API keys?

© UMHackathon 2026

Email: umhackathon@um.edu.my

- Mitigation: All sensitive credentials are stored in Environment Variables (.env) and never hardcoded in the source code or stored in client-side local storage.